

ADAPT

Where Does an AI Learn a New Rule?

One-page concept summary | self-contained briefing

THE QUESTION

When a model encounters an unseen task, watches a few demonstrations, and answers a new query, **where does the new rule live?** ADAPT turns that question into a controlled experiment so a learner can see what changes and why, without claiming that one mechanism explains all AI systems.

THREE ADAPTATION SUBSTRATES

MECHANISM	RULE STORAGE	dTheta	dState
Frozen / context	Examples remain in context.	0	0
Parameter TTA	Gradient-updated weights.	>0	0
State adaptation	Adaptive internal register.	0	>0
Learned state / TTT	Learned adaptive state.	0	>0

WHY THE DISTINCTION MATTERS

Context, parameter updates, and adaptive state can all improve predictions, but they pay different costs. State adaptation keeps persistent parameters fixed while an internal state changes after each demonstration. That state has finite capacity: compress it too aggressively and the statistics needed for the query disappear; reuse a shared state across tasks and new rules can overwrite old ones. ADAPT exposes both effects through capacity and $A \rightarrow B \rightarrow A$ interference experiments.

CAUSAL EVIDENCE

The intervention experiment perturbs the adapted state, collapsing prediction accuracy to roughly 0.03-0.10, and then restores it, recovering 1.00 in the tested setup. That makes the state causally relevant to the current prediction rather than merely a visual trace.

CONTROLLED RESULTS

On 200 paired unseen linear tasks, frozen is 0.11 while state adaptation is 1.00 with $d\Theta = 0$ and $dS > 0$. On the controlled symbolic family, frozen is 0.00 and state adaptation is 1.00. The learned recurrent checkpoint remains near frozen at 0.09 vs 0.11 under the tested meta-training budget; this honest negative is retained.

BDH / BDH-CQ FRONTIER

BDH (Kosowski et al., 2025) places working memory in a synaptic state updated Hebbian-style and separate from persistent parameters. BDH-CQ (Engdahl et al., 2026) combines recurrent memory updates with iterative latent reasoning. ADAPT is **not** an official implementation of either system; it is a transparent controlled analogue of the shared design pressure: **demonstrations write adaptive memory, and the query is solved from that adapted substrate.**

EVIDENCE AND LIMITS

Learner-facing results are either computed live or explicitly labelled as precomputed, published, synthetic, or illustrative. The grid environment is an ARC-like proxy, not ARC-AGI. BDH / BDH-CQ headline numbers remain published claims unless independently reproduced. The learning study is preliminary at $n=4$; the frozen protocol targets 8-15 participants.

TAKEAWAY

Inference-time adaptation is not one mechanism. The substrate that stores a new rule determines what the system can remember, what computation it pays for, and how it fails.

Selected primary references: Kosowski et al. (2025), arXiv:2509.26507; Engdahl et al. (2026), arXiv:2608.09888; Akyürek et al. (2025), ICML; Sun et al. (2025), ICML; von Oswald et al. (2023), ICML. Repository: github.com/Adithkp03/ADAPT